An audit of the immrep25 unseen-peptide benchmark

Signal-to-noise of CDR3 homology and V/J chain pairing across the positive set

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Abstract

We ask whether the *positive* TCR–peptide assignments in the immrep25 unseen-peptide benchmark carry epitope-specific signal, or only noise and publicity. Two biologically-grounded probes—within- vs. between-epitope CDR3 *homology* and per-epitope V/J *chain pairing*—are evaluated one-vs-many per epitope and aggregated across epitopes, then calibrated against cohorts of known quality (TCRvdb experimentally verified / refuted; VDJdb high- / low-reference-support) and three controls that fix the noise floor at S/N=1: two OLGA controls (raw random and generation-probability–matched) and a real post-thymic-selection bulk repertoire (AIRR). On both probes and both chains the immrep25 positives sit just above that floor and one to three orders of magnitude below every quality cohort; their weak homology is fully explained by public TCRs (it collapses to the floor once public TCRs are removed) and their inter-chain pairing shows essentially no coupling. We conclude the positive labels carry no meaningful epitope-specific signal beyond noise and publicity.

1 Data and design

Cohorts. Every cohort is reduced to a common per-chain schema (one CDR3 with its V/J and epitope) and a paired schema ($\mathrm{CDR3}_\alpha, V_\alpha, J_\alpha, \mathrm{CDR3}_\beta, V_\beta, J_\beta$, epitope). The six cohorts span the expected signal-to-noise (S/N) range:

cohort	definition	role
TCRvdb true	TCRvdb, $p_{\text{adj}} < 10^{-5}$	experimentally verified (top)
VDJdb HQ	VDJdb human MHC-I, ≥ 2 references per (TCR, epitope)	strongly supported
VDJdb LQ	VDJdb human MHC-I, single reference	weakly supported
TCRvdb false	TCRvdb, $p_{\text{adj}} \geq 10^{-5}$	experimentally refuted
immrep25 pos	immrep25 `label`=1	**system under test**
AIRR random	pooled real repertoire (isalgo/airr_control), random labels	real-data floor
AIRR non-random	20 largest SRA donors; epitope = a donor's top-50 clonotypes	structured real control
OLGA random	raw OLGA generation, random labels	noise floor

Cohorts draw on VDJdb [1], the functionally-validated TCRvdb/MATCHMAKERS database [2], the immrep25 unseen-peptide benchmark [3], and an OLGA control built from the TCR generation model of Murugan et al. [4]. The immrep25 test set covers 20 nine-mer peptides that are *absent from VDJdb*, with 50 positives (and 450 negatives) each; only positives are audited. VDJdb HQ is defined per chain (TRA and TRB separately) as a (TCR, epitope) supported by ≥ 2 distinct references, with a paired complex counted HQ if *either* chain qualifies. TCRvdb true/false are dominated by the same two epitopes (YLQPRTFLL, GLCTLVAML), giving an epitope-matched verified-vs-refuted contrast.

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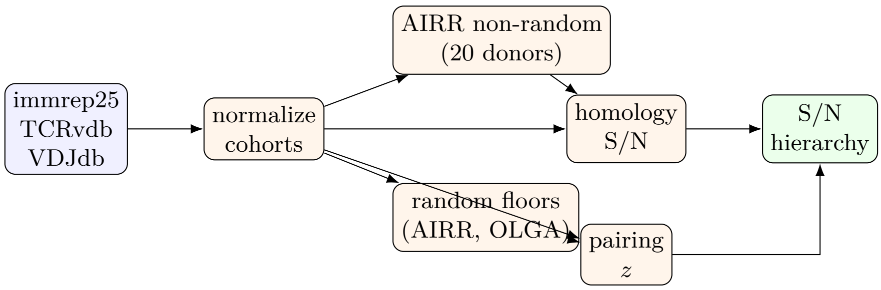
One-vs-many. Every metric is computed *per epitope* against all others (each epitope with ≥ 30 records vs. the rest of its cohort) and then aggregated to a dataset value—geometric mean $\pm$ 95% CI for homology S/N, mean $\pm$ 95% CI for pairing—across epitopes. The signal therefore attaches to individual epitopes and the reported error bars are the epitope-to-epitope spread (n = number of qualifying epitopes); no bootstrap resampling is used.

A structured real-data control. Random synthetic labels give S/N=1 by construction, so we add a control whose epitope grouping is *real and structured*: the **AIRR non-random** cohort. From the isalgo/airr_benchmark SRA collection we take the 20 largest samples that carry ≥ 50 TRA and ≥ 50 TRB clonotypes; each *donor* contributes its top 50 TRA and top 50 TRB clonotypes (by read count) as one virtual epitope. This mimics how a real epitope-specific dataset is built (expanded clones per sample), but the "epitopes" are individual donors, not antigens. Two random-label floors (OLGA random and the pooled AIRR random) are kept for reference. The 20 donors are listed in `results/airr_donors.tsv`.

Pipeline.



2 Homology test

Metric. For a single chain we score each epitope one-vs-many. Only equal-length CDR3s can share a Hamming distance, so pairs are counted within length bins. For epitope e (N_e TCRs) against the other $N - N_e$ TCRs of its cohort,

$$P_{\text{self}} = \binom{N_e}{2}, \qquad P_{\text{non}} = N_e\,(N - N_e)$$

(equal-length pairs); with $m_{\leq d}/l_{\leq d}$ the within-/against-rest pairs at Hamming $\leq d$ and an exposure-proportional pseudocount a (so S/N$\to$1 under no enrichment),

$$\text{S/N}_e(d) = \frac{(m_{\leq d} + a)/P_{\text{self}}}{\big(l_{\leq d} + a\,P_{\text{non}}/P_{\text{self}}\big)/P_{\text{non}}}.$$

The dataset value is the geometric mean of S/N$_e$ over epitopes ($\pm$95% CI); the error bars are thus the epitope-to-epitope spread. d=0 isolates publicity (identical CDR3s); d=1 is the substitution-neighbour convergence signal exploited by curated specificity databases [1]; signal $\Rightarrow$ S/N $\gg 1$, noise $\Rightarrow$ S/N ≈ 1.

Result. Figure 1 orders the cohorts: VDJdb-HQ > TCRvdb-true > TCRvdb-false > VDJdb-LQ $\gg$ immrep25 > OLGA. On TCRβ the verified cohorts reach S/N = 1756 (VDJdb-HQ) and 496 (TCRvdb-true), whereas immrep25 positives give only 2.69 and *every* control sits at the floor—including the structured **AIRR non-random** control (1.04), whose 20 epitopes are the top-50 expanded clonotypes of 20 real donor repertoires. Even this realistic construction does not reproduce immrep25's within-epitope homology: each donor's top clones are private expansions (no β clonotype is shared between donors), so the "epitopes" carry no convergent structure. immrep25's excess is therefore specific to its epitope→TCR assignments (i.e. publicity, §4), not to abundance or expansion in real data (AIRR random 1.00, OLGA random 0.969). The same holds on TCRα (immrep25 3.88). immrep25's CI excludes 1 but lies one to three orders of magnitude below every quality cohort. As the substitution radius grows (Fig. 2) the verified cohorts stay high while immrep25 and all controls remain near the floor across $d = 1, 2, 3$.

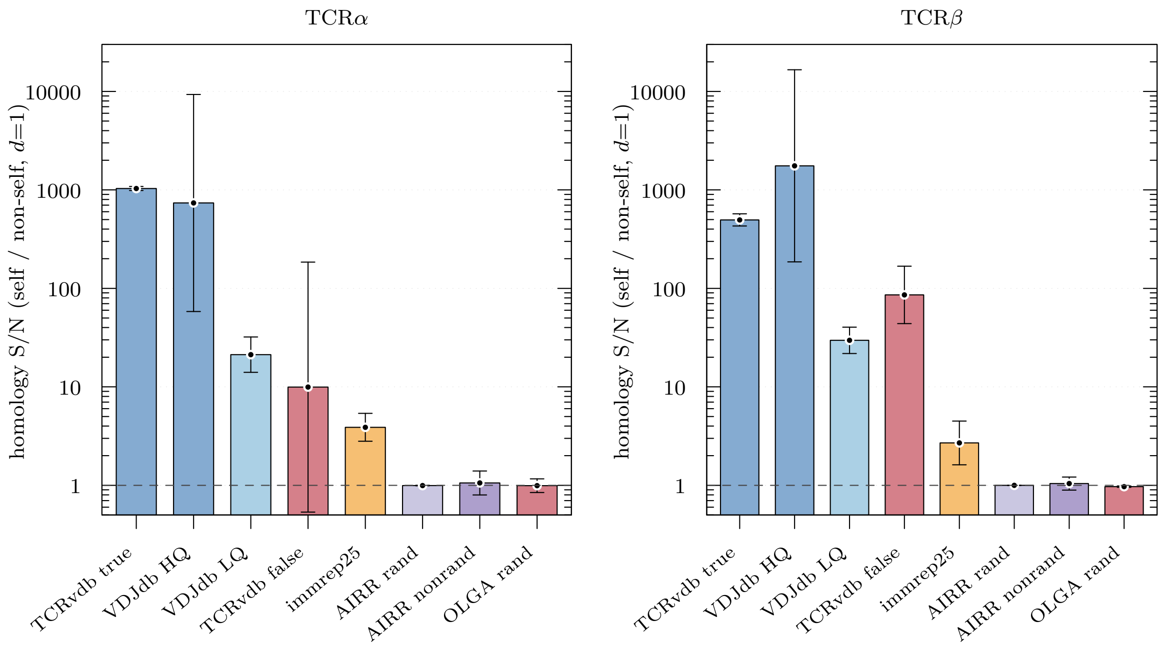


Figure 1: Homology S/N (d=1) per cohort, TCRα and TCRβ (one-vs-many: each epitope with ≥ 30 records vs. the rest; bar = geometric mean over epitopes, whiskers = 95% CI = between-epitope spread). Dashed line: noise floor S/N=1.

By-epitope breakdown. Figure 3 plots the within-epitope neighbour rate for every epitope with $n \geq 30$ (one point per epitope; boxplots show the per-cohort quartiles); the vertical gap to each cohort's non-self rate (red bar) is the S/N. immrep25's points hug its non-self line, whereas the verified cohorts sit orders of magnitude above. Table 1 lists all 20 immrep25 epitopes: in total they contribute only 54 (TCRβ) / 53 (TCRα) within-epitope Hamming-1 neighbours — versus 0 / 2 for the OLGA random control and tens of thousands for the VDJdb cohorts — and even that handful is concentrated in a few epitopes (e.g. YLFNADIWI) and is entirely public (§4). By contrast the AIRR non-random donors contribute essentially no within-epitope neighbours (2 TCRβ), because each donor's expanded clones are private rather than epitope-convergent.

Graph view. The S/N ratio is the within- vs. between-epitope edge-density of the homology graph (nodes=CDR3, edges=Hamming≤1; all epitopes with ≥ 30 records). Figure 4 renders it (graphviz `sfdp` layout): the verified cohorts form dense epitope-coloured cliques, while immrep25 and all controls are near-edgeless point clouds with no epitope-coloured clustering.

3 Pairing test

Metric. Genome-wide α/β pairing is close to random [5], but given an epitope the V/J usage may be constrained. For every epitope with ≥ 30 paired records we measure two signals as the *excess over a permutation null*, in bits, then average over epitopes (±95% CI):

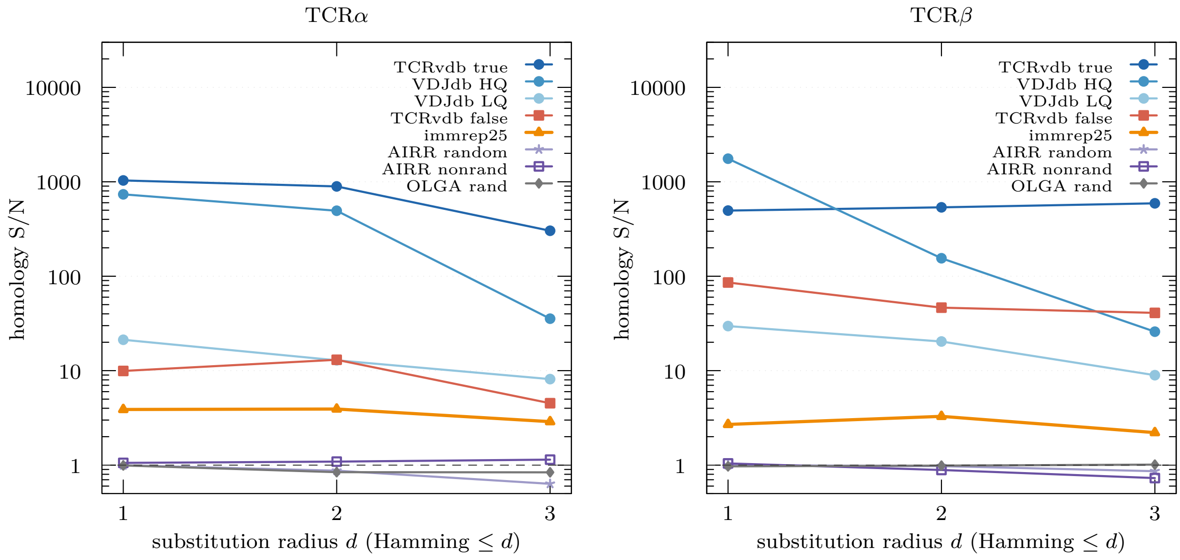


Figure 2: Homology S/N vs. substitution radius $d = 1, 2, 3$ (one-vs-many geometric mean). Verified cohorts stay high; immrep25 and all controls track the floor.

- **gene-usage bias**: $\overline{\mathrm{KL}}(P_e \,\|\, Q)$ over axes $V\alpha, J\alpha, V\beta, J\beta$ (P_e = within-epitope gene distribution, Q = cohort background) minus its mean under size-matched random draws from Q.
- **inter-chain MI**: $I((V\alpha, J\alpha); (V\beta, J\beta))$ within epitope (Miller–Madow) minus its mean under $\alpha \leftrightarrow \beta$ shuffles (which keep the marginals but destroy coupling).

Both are 0 under no signal and grow (in bits) with genuine per-epitope structure. Writing X for either statistic and $\{X_{\mathrm{null}}^{(k)}\}$ for its K permutation replicates, the excess is $\Delta X_e = X_e - \overline{X_{\mathrm{null}}}$ and the standardized effect is

$$z_e = \frac{X_e - \overline{X_{\mathrm{null}}}}{\mathrm{sd}(X_{\mathrm{null}})}.$$

We report the excess ΔX (in bits) rather than z, because for small epitopes the null variance can collapse (sd $\to 0$) and inflate z; the excess is stable and keeps the natural unit.

Result. Figure 5 shows both. The verified cohorts carry substantial gene-usage bias (VDJdb-HQ 1.62 bits) and real inter-chain coupling (TCRvdb-true 1.04 bits). immrep25 positives are near the floor on both: gene-usage bias 0.203 bits and, decisively, an inter-chain coupling of only 0.047 bits—an order of magnitude below every quality cohort and barely above the AIRR and OLGA controls (0.001 / −0.000 bits). The residual gene-usage bias is consistent with the public, HLA-restricted composition of the positives rather than epitope-specific pairing.

4 Publicity control

The one alternative to "pure noise" is *publicity*: immrep25 might reuse public TCRs whose chance homology mimics epitope signal. Although only 0.3% of positives share *both* chains exactly with VDJdb, allowing a single substitution and either chain, 74.5% of immrep25 positives are within Hamming ≤ 1 of a known VDJdb/TCRvdb TCR (61.5% on α, 31.2% on β)—the α chain being especially germline-public. Consistently, the positives are enriched for high generation probability relative to raw OLGA sampling (Fig. 6), i.e. they are "easy-to-generate", public-leaning sequences; this is exactly why the random control must be P_{gen}-matched.

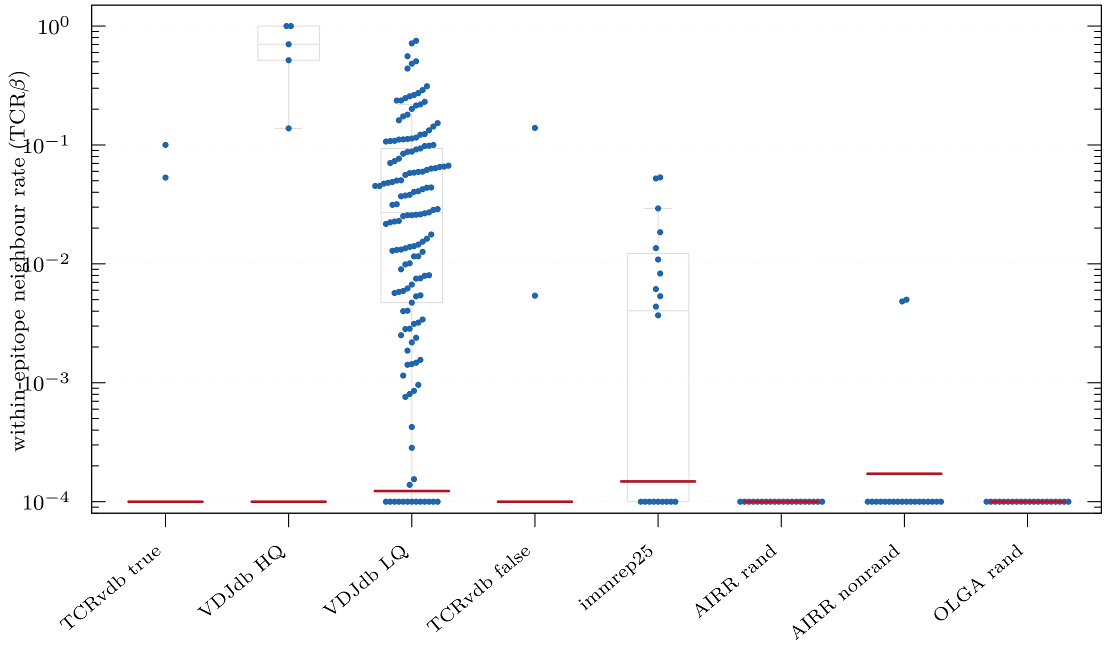


Figure 3: Per-epitope within-epitope neighbour rate (TCRβ, Hamming$\leq$ 1; each point is one epitope, $n \geq 30$; boxplots = per-cohort quartiles). Red bars = cohort pooled cross-epitope (non-self) rate; the vertical gap is the homology S/N.

Removing public positives collapses the (already weak) homology to the noise floor: one-vs-many S/N falls from 2.69 to 1.00 on TCRβ and from 3.88 to 1.37 on TCRα (Fig. 7). The residual signal was therefore publicity, not epitope-specific convergence.

5 Neighbourhood pgen (generation degree)

Point P_{gen} controls how likely a single CDR3 is to be generated, but chance homology depends on how many likely *neighbours* a sequence has. Following mirpy [6], we quantify this with the *1-mm neighbourhood pgen*—the generation probability of all CDR3s within Hamming distance 1, computed exactly with OLGA's wildcard identity (`compute_hamming_dist_1_pgen`; terminal anchor positions skipped). Figure 8 shows the per-chain distributions.

Degree varies widely, yet homology S/N stays at the floor. The cohorts span a broad range of generation degree. The pooled AIRR random repertoire, sampled over *unique* clonotypes, sits lowest (its diversity is dominated by low-pgen singletons); raw OLGA generation is intermediate; and immrep25 together with the AIRR non-random donor clones sit highest—immrep25 positives are public-leaning, high-P_{gen} sequences (Fig. 6), and a donor's most expanded clonotypes are likewise abundant and public. Despite this spread, *every* control sits at homology S/N≈1 (Fig. 1): the S/N is normalised against a random re-partition of each cohort, so it measures epitope *specificity*, not absolute degree. Generation degree and epitope-specific convergence are distinct axes—immrep25 is elevated on neither beyond publicity.

6 Conclusion

Across both probes and both chains the calibration cohorts recover the expected order, and the immrep25 positives fall at the bottom, just above the real-repertoire (AIRR) and OLGA floors and one to three orders of magnitude below every quality cohort (Fig. 9). Their weak homology

epitope	n	β nbrs	α nbrs	β S/N
`YLFNADIWI`	50	20	1	29.35
`MTDYDYLEV`	50	13	3	27.00
`TVYPYGTSL`	50	5	2	11.00
`ILHTHVPEV`	50	5	7	9.11
`YMFYDGYDV`	50	3	2	6.28
`SQFNWTIYL`	50	2	1	5.00
`REDDYSVWL`	50	2	1	4.52
`SEESAFYVL`	50	1	2	3.00
`FLDCKSYIL`	50	1	4	2.86
`GEVLACYAL`	50	1	1	2.83
`FEAQPGALL`	50	1	3	2.59
`MEMPDYLLL`	50	0	3	1.00
`HLDDYPYLM`	50	0	1	1.00
`KLYPFLWFA`	50	0	11	1.00
`GENALTYAL`	50	0	3	1.00
`YENGSTPVL`	50	0	1	0.95
`YEDGVIFYL`	50	0	5	0.91
`ILMHATYFL`	50	0	1	0.90
`MENWSALEL`	50	0	0	0.88
`YESYIPGAL`	50	0	1	0.83
immrep25 (geom. mean)	1000	54	53	2.69
AIRR random	1000	0	0	1.00
AIRR non-random	1000	2	11	1.04
OLGA random	1000	0	2	0.97

Table 1: immrep25 positives by epitope (n TCRs) and their count of within-epitope Hamming-1 CDR3 neighbours per chain and β S/N, versus the structured/random controls. Sorted by β S/N.

is contributed by public TCRs and collapses to the floor when those are removed; their inter-chain pairing shows essentially no coupling. We therefore find *no meaningful epitope-specific signal—in homology or chain pairing—among the immrep25 positives beyond noise and publicity*, consistent with the benchmark's own finding that models do not exceed random guessing on unseen peptides [3].

Relation to reported prediction scores. The earlier IMMREP23 challenge [7] was built largely on curated VDJdb-quality data and reported that specificity can be predicted for its *seen* peptides with useful accuracy (median $\mathrm{AUC}_{0.1} \gtrsim 0.7$)—consistent with our finding that the VDJdb cohorts carry strong homology and pairing signal (Figs. 1, 5). IMMREP25, in contrast, found no method exceeded a best macro-$\mathrm{AUC}_{0.1}$ of 0.60 on *unseen* peptides [3]. Our audit accounts for the gap without running any predictor: the unseen-peptide positives carry no epitope-specific signal beyond publicity, so a model has nothing epitope-specific to latch onto and can only exploit similarity to previously seen, VDJdb-like TCRs—which novel epitopes, by construction, do not provide.

Caveats. TCRvdb true/false separate less than expected because p_{adj} scores differential enrichment in one screen rather than absence of specificity, so "false" still contains epitope-associated TCRs; and TCRvdb and VDJdb-HQ rest on only 2 and 5 epitopes respectively, so their absolute S/N is uncertain (wide CIs). Neither affects immrep25, whose 20 epitopes place it just above the real-repertoire (AIRR) and OLGA floors on every probe. Reproduce with the fetch scripts (`fetch_vdjdb.sh`, `fetch_airr.sh`, `gen_olga_pool.sh`), then `build_olga.py`, `build_airr.py`, `run_audit.py`, `make`.

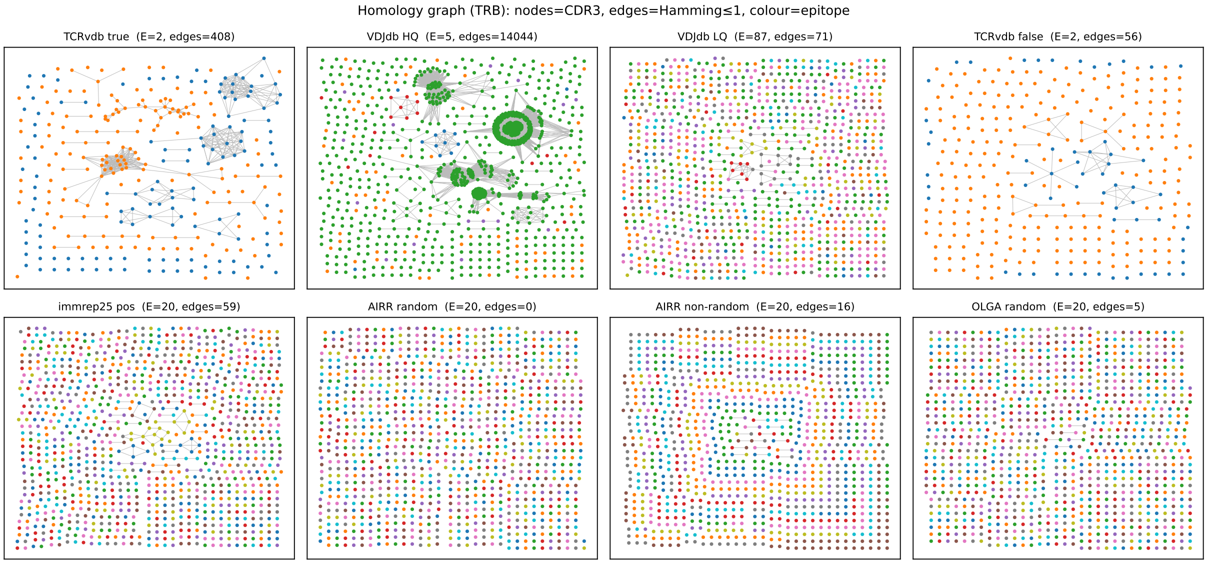


Figure 4: TCRβ homology graph per cohort (all epitopes $\geq$ 30 records). Verified cohorts form dense epitope-coloured cliques; immrep25 and all controls are near-edgeless.

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[1] Mikhail Shugay, Dmitriy V Bagaev, Ivan V Zvyagin, Renske M Vroomans, Jeremy Chase Crawford, Garry Dolton, Ekaterina A Komech, Anastasiya L Sycheva, Anna E Koneva, Evgeniy S Egorov, Alexey V Eliseev, Ewald Van Dyk, Pradyot Dash, Meriem Attaf, Cristina Rius, Kristin Ladell, James E McLaren, Katherine K Matthews, E Bridie Clemens, Daniel C Douek, Fabio Luciani, Debbie van Baarle, Katherine Kedzierska, Can Kesmir, Paul G Thomas, David A Price, Andrew K Sewell, and Dmitriy M Chudakov. Vdjdb: a curated database of t-cell receptor sequences with known antigen specificity. *Nucleic Acids Research*, 46(D1):D419–D427, September 2017.

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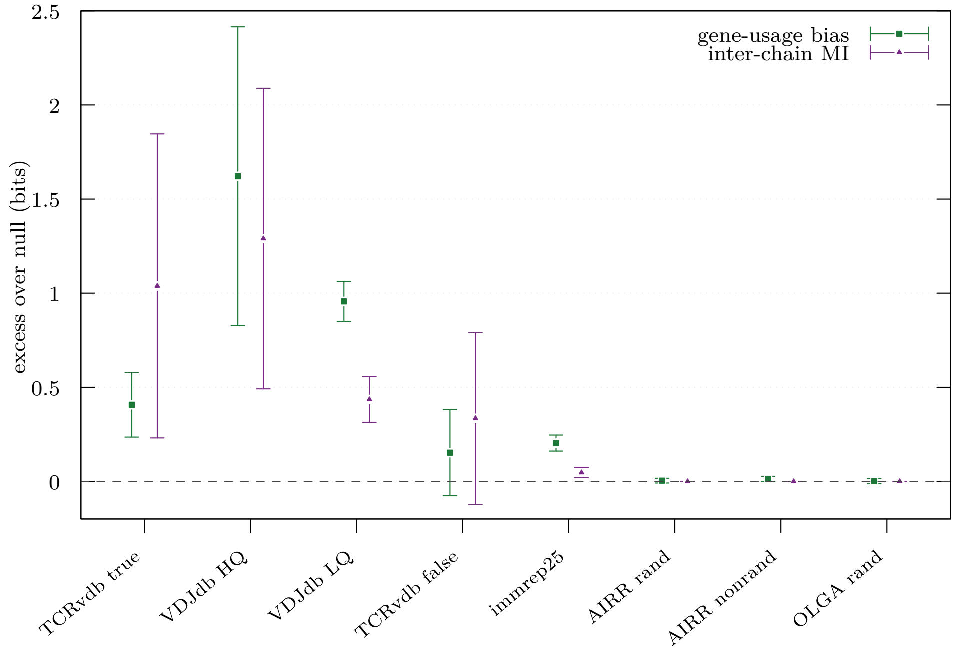

Figure 5: Pairing signals per cohort (one-vs-many over all epitopes with $\geq$ 30 records): per-epitope excess over the permutation null, in bits (mean $\pm$ 95% CI). Squares = gene-usage bias, triangles = inter-chain MI. Dashed line = 0 (random). immrep25 and all controls sit at the random line, especially for MI.

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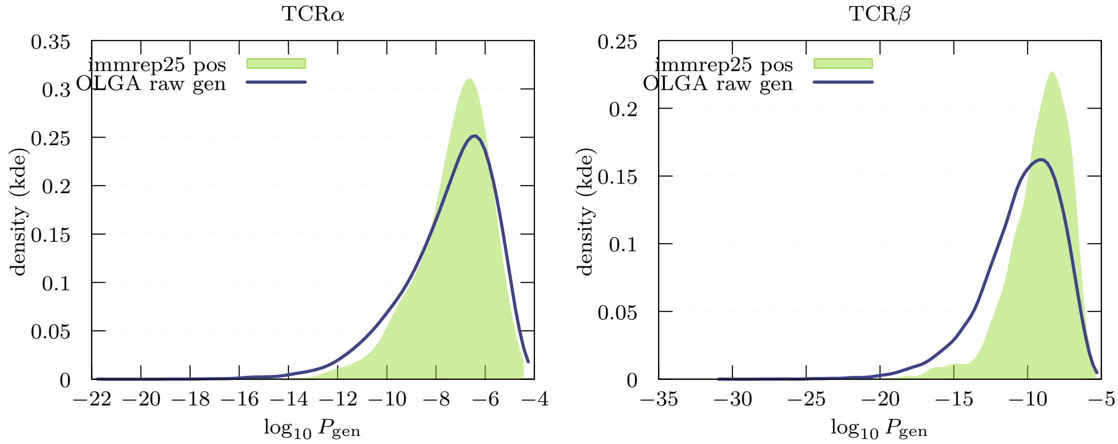

Figure 6: Per-chain $\log_{10} P_{\text{gen}}$ (kernel density): immrep25 positives (filled) vs. raw OLGA generation (line). Positives skew to higher P_{gen} (public-leaning), motivating the P_{gen}-matched control.

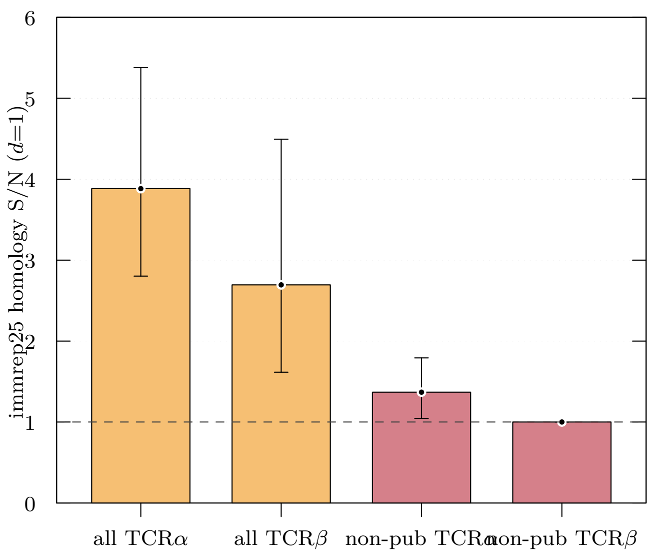

Figure 7: immrep25 homology S/N (d=1) for all positives vs. non-public positives (Hamming ≤ 1 to references removed). Removing public TCRs drives S/N to the noise floor (dashed line).

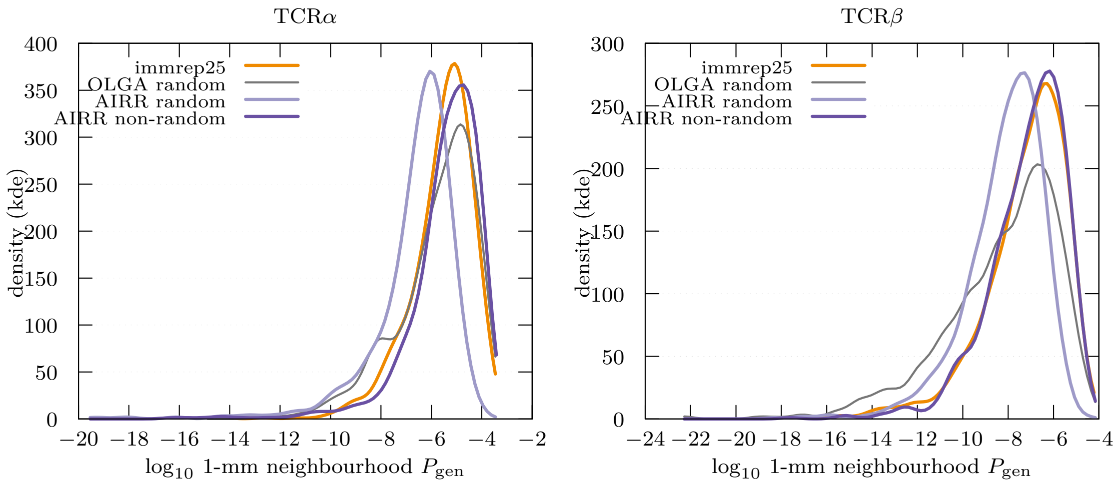

Figure 8: 1-mm neighbourhood P_{gen} (kernel density) per chain. immrep25 and the abundant AIRR non-random donor clones sit at high degree, the pooled AIRR random repertoire at low degree, OLGA random in between—yet all controls sit at homology S/N≈1 (Fig. 1).

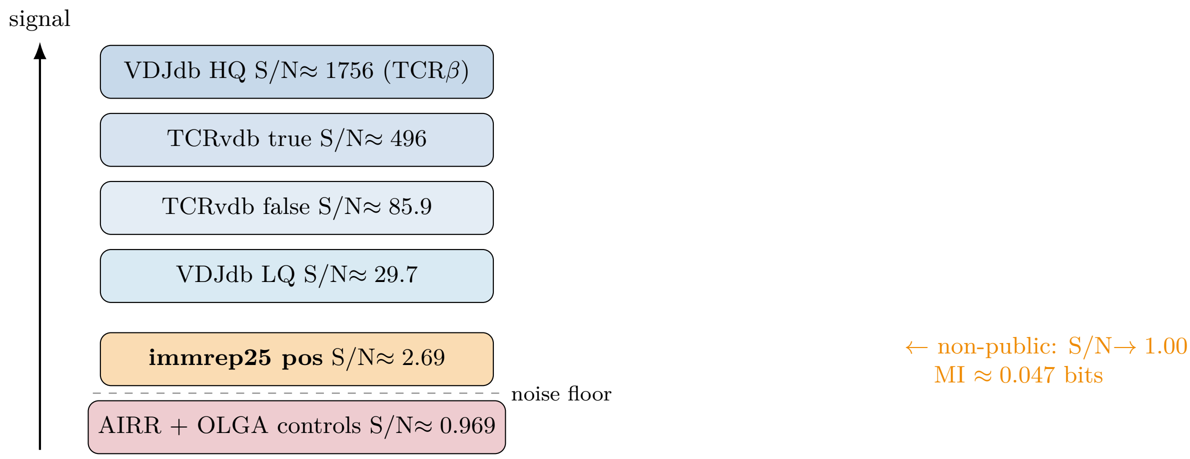

Figure 9: Signal-to-noise ladder (TCRβ homology, one-vs-many). The immrep25 positives sit one rung above the random floor; stripping public TCRs lowers them onto it.